

Physics based Machine Learning Model to Predict how real gases behave

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Industry Partner: Future Energy Exports

Introduction & Background

Hydrogen is vital for achieving net zero emissions by 2050, especially in transport, aviation and energy sector. However, existing real-gas Equations of State (EoS) models either poorly capture its behaviour or are computationally expensive under high-pressure and cryogenic conditions. This project uses machine learning framework to develop an accurate and computationally efficient EoS for safe hydrogen deployment.

Objective

- Deliver Artificial Neural Network model to predict thermodynamic properties such as speed of sound and heat capacities.
- Validate the model under cryogenic conditions.

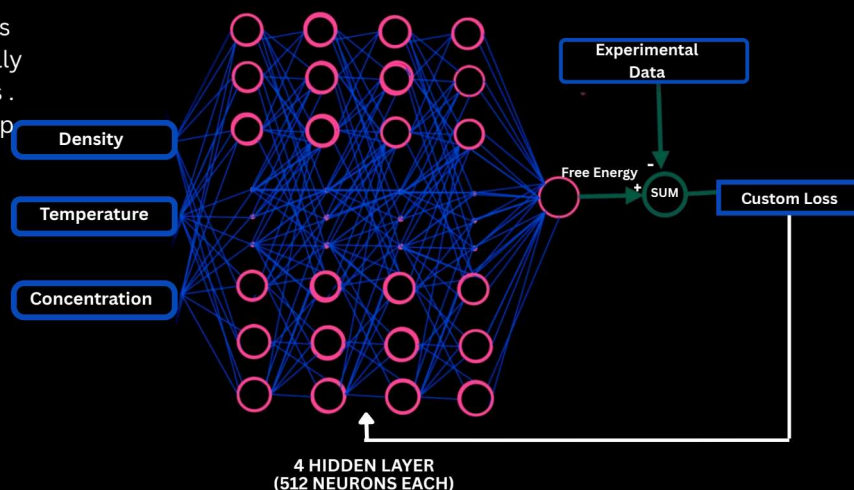
Limitations of present Equations of State

- Computationally expensive
- Not GPU-friendly
- Not reliable under cryogenic extremes
- Limited or missing data near critical regions

Method

- Use available experimental data to train a predictive model for the Helmholtz Free Energy (F).
- Derive thermodynamic properties — such as speed of sound, specific heats, and enthalpy — from the predicted Helmholtz free energy using automatic differentiation.

Model Architecture

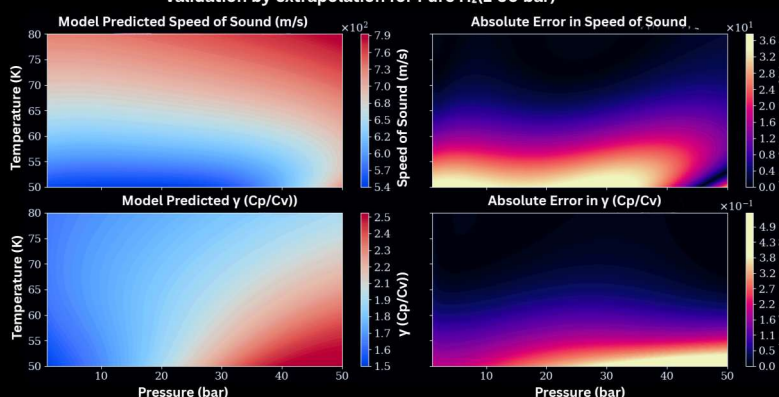


Prediction Variables

- Primary: Helmholtz Free Energy, $F = U - TS$, where U is the internal energy and TS is the energy unavailable to do work.
- Secondary: Pressure, internal energy, entropy, C_v , C_p and speed of sound.

Results

Validation by extrapolation for Pure H₂(1-50 bar)



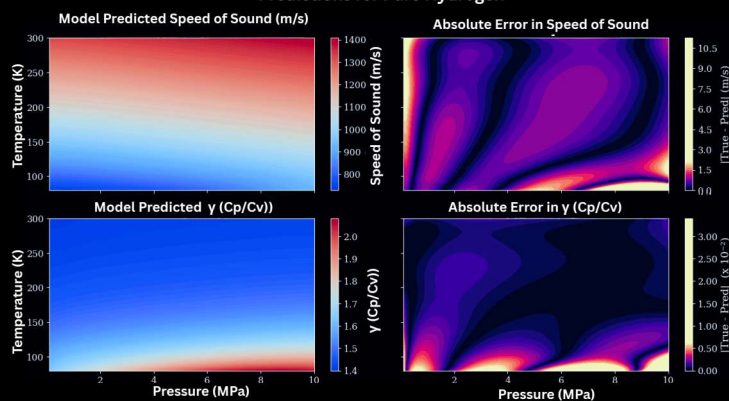
Test & Validation

- Compare model predictions with experimental data.
- Evaluate model performance outside the training region, particularly at cryogenic temperatures, and verify whether the model converges to the ideal-gas limit.

Future Work

- Generalize the methodology to extend the approach to other gas mixtures critical for industrial applications.
- Integrate the trained model with a CFD solver to simulate real-gas flows and optimize engineering designs.

Predictions for Pure Hydrogen



Achievements

- Demonstrated that a physics-based ML approach can be used as a predictive tool to compute thermodynamic properties.
- Achieved close agreement with experimental data outside the trained region within the same phase.
- This method has been extended to real-gas mixtures and can serve as a GPU-friendly workflow for CFD integration.